

Statistics for Social Science Research

Lecture 6

07 September 2026

Topics for today

- The t Test for Two Independent Samples (Chapter 10, Statistics for the Behavioral Sciences)
- The t Test for Two Related Samples (Chapter 11, Statistics for the Behavioral Sciences)

(The slides in this power point are partly taken from the Power Point Lecture Slides to accompany the textbook “Essentials of Statistics for the Behavioral Sciences” by Frederick J Gravetter and Larry B. Wallnau.)

Chapter 10

The t Test for Two Independent Samples

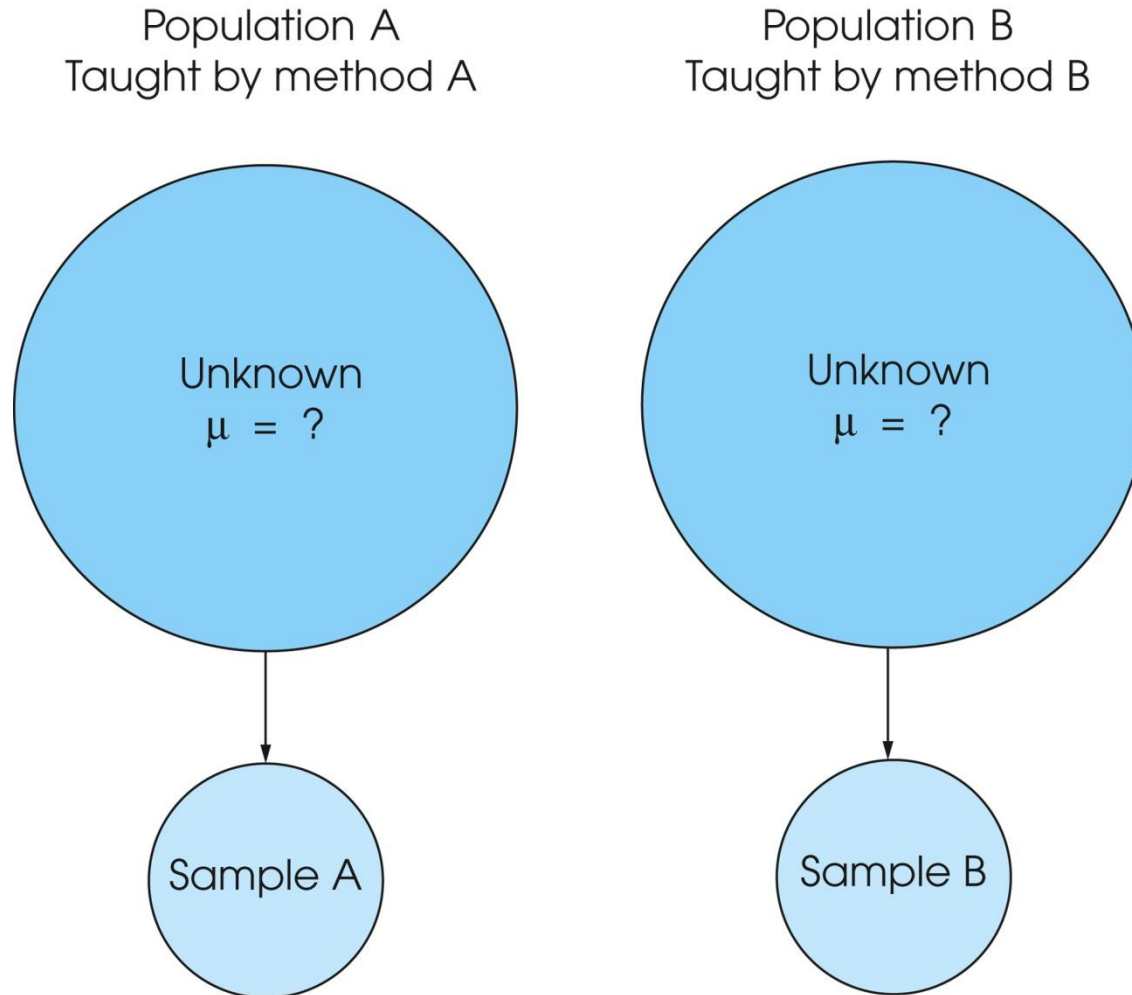
10.1 Independent-Measures Design Introduction

- Most research studies compare two (or more) sets of data
 - Data from two completely different, independent participant groups (an independent-measures or between-subjects design)
 - Data from the same or related participant group(s) (a within-subjects or repeated-measures design)

10.1 Independent-Measures Design Introduction (continued)

- Computational procedures are considerably different for the two designs
- Each design has different strengths and weaknesses
- Consequently, only between-subjects designs are considered in this chapter; repeated-measures designs will be reserved for discussion in Chapter 11

Figure 10.1 Independent-Measures Research Design



10.2 Independent-Measures Design t Statistic

- Null hypothesis for independent-measures test

$$H_0 : \mu_1 - \mu_2 = 0$$

- Alternative hypothesis for the independent-measures test

$$H_1 : \mu_1 - \mu_2 \neq 0$$

Independent-Measures Hypothesis Test Formulas

- Basic structure of the t statistic

$$t = \frac{(M_1 - M_2) - (\mu_1 - \mu_2)}{S_{(M_1 - M_2)}}$$

- $t = [(\text{sample statistic}) - (\text{hypothesized population parameter})]$ divided by the estimated standard error
- Independent-measures t test

Estimated standard error

- Measure of standard or average distance between sample statistic ($M_1 - M_2$) and the population parameter
- How much difference it is reasonable to expect between two sample means if the null hypothesis is true (equation 10.1)

$$S_{(M_1 - M_2)} = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

Pooled Variance

- Equation 10.1 shows standard error concept but is unbiased only if $n_1 = n_2$
- Pooled variance (s_p^2 provides an unbiased basis for calculating the standard error)

$$s_p^2 = \frac{SS_1 + SS_2}{df_1 + df_2}$$

Degrees of freedom

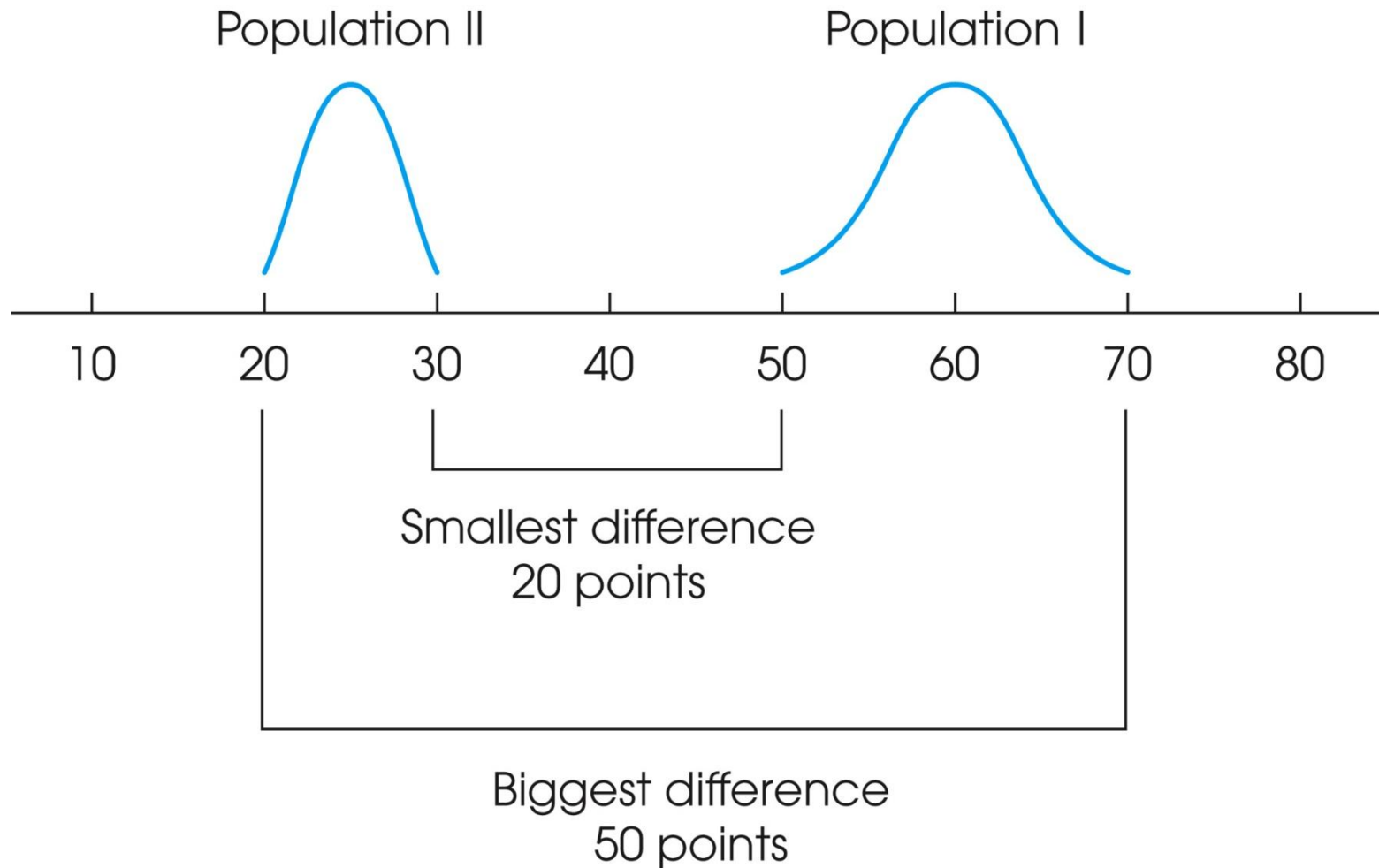
- Degrees of freedom (df) for t statistic is df for first sample + df for second sample

$$df = df_1 + df_2 = (n_1 - 1) + (n_2 - 1)$$

Note: this term is the same as the denominator of the pooled variance

Figure 10.2

Two population distributions

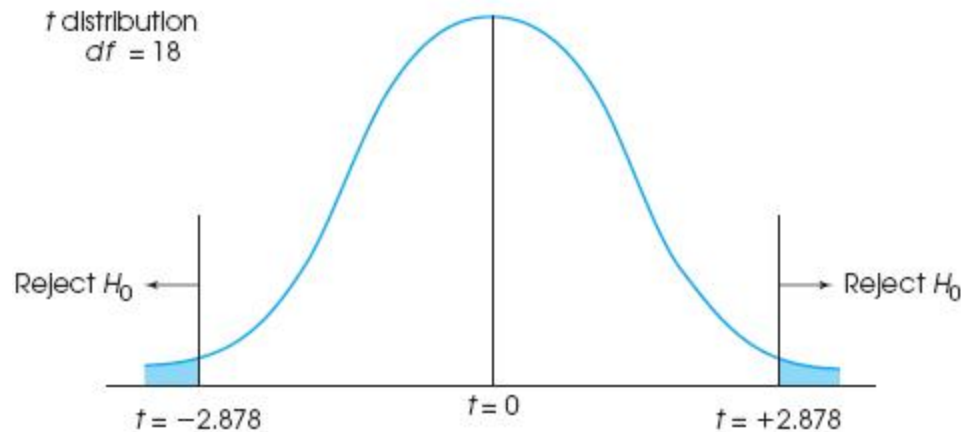


Estimated Standard Error for the Difference Between Two Means

The estimated standard error of $M_1 - M_2$ is then calculated using the pooled variance estimate

$$s_{(M_1 - M_2)} = \sqrt{\frac{s_p^2}{n_1} + \frac{s_p^2}{n_2}}$$

Figure 10.3 Critical Region for Example 10.1 ($df = 18$; $\alpha = .01$)



Learning Check

- Which combination of factors is most likely to produce a significant value for an independent-measures t statistic?

A

- a small mean difference and small sample variances

B

- a large mean difference and large sample variances

C

- a small mean difference and large sample variances

D

- a large mean difference and small sample variances

Learning Check

- Decide if each of the following statements is True or False

T/F

- If both samples have $n = 10$, the independent-measures t statistic will have $df = 19$

T/F

- For an independent-measures t statistic, the estimated standard error measures how much difference is reasonable to expect between the sample means if there is no treatment effect

Measuring Effect Size

- If the null hypothesis is rejected, the size of the effect should be determined using *either*
- Cohen's d

$$\text{estimated } d = \frac{\text{estimated mean difference}}{\text{estimated standard deviation}} = \frac{M_1 - M_2}{\sqrt{s_p^2}}$$

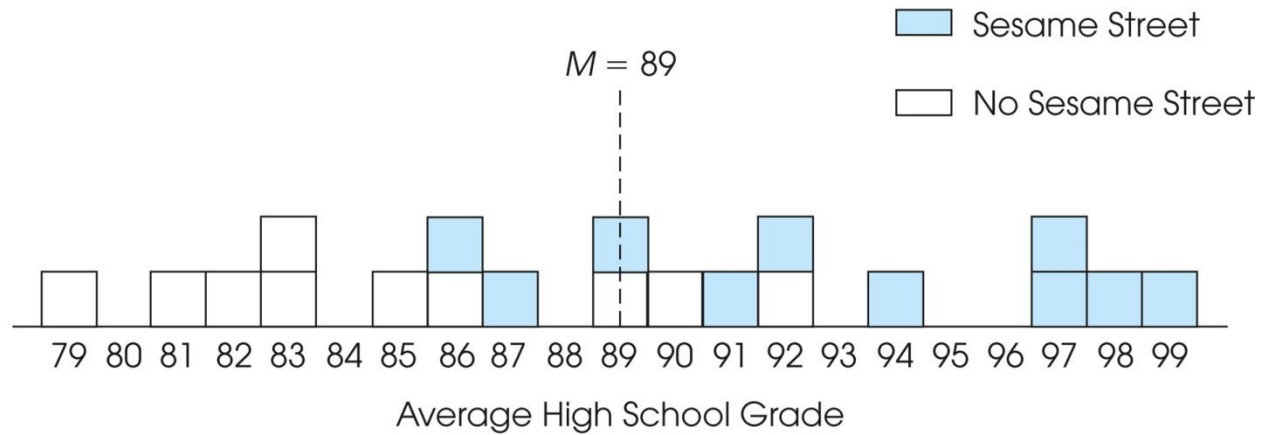
- *or* Percentage of variance explained

$$r^2 = \frac{t^2}{t^2 + df}$$

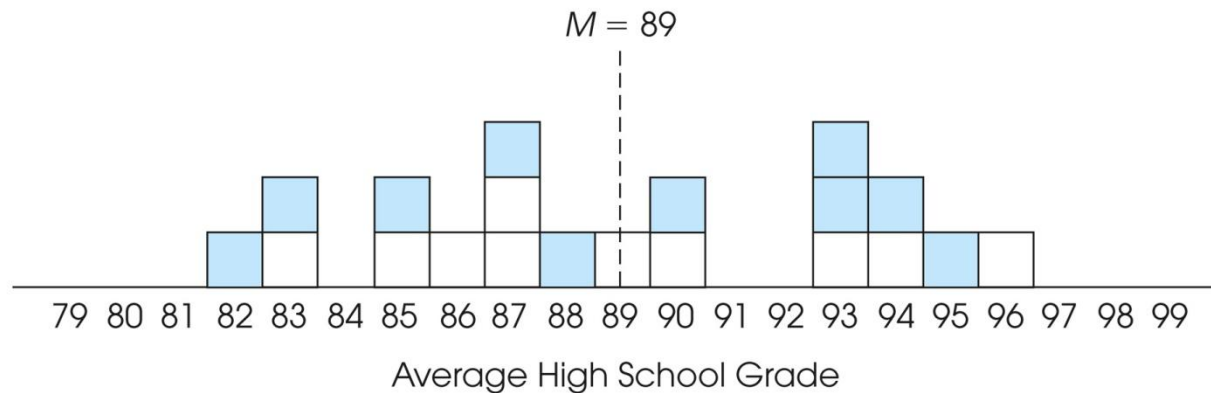
Figure 10.4

Scores from Example 10.1

(a) Original scores including the treatment effect



(b) Adjusted scores after the treatment effect is removed



Confidence Intervals for Estimating $\mu_1 - \mu_2$

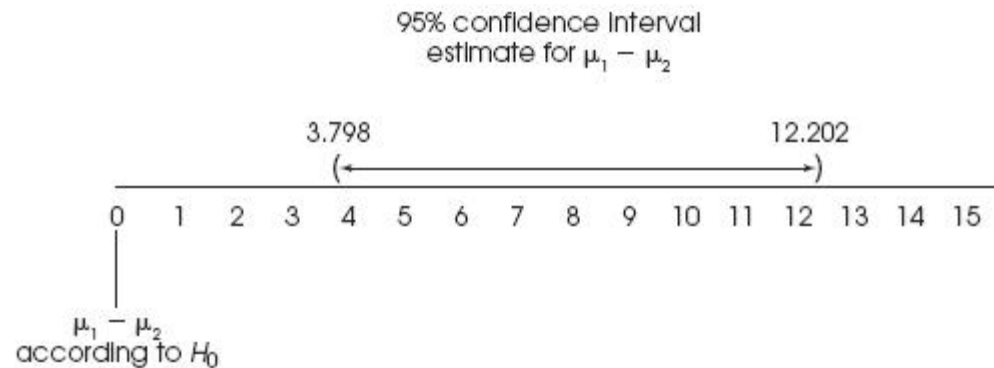
- Difference $M_1 - M_2$ is used to estimate the population mean difference
- t equation is solved for unknown $(\mu_1 - \mu_2)$

$$\mu_1 - \mu_2 = M_1 - M_2 \pm ts_{(M_1 - M_2)}$$

Confidence Intervals and Hypothesis Tests

- Estimation can provide an indication of the size of the treatment effect
- Estimation can provide an indication of the significance of the effect
- If the interval contain zero, then it is not a significant effect
- If the interval does NOT contain zero, the treatment effect was significant

Figure 10.5 95% Confidence Interval from Example 10.3



In the Literature

- Report whether the difference between the two groups was significant or not
- Report descriptive statistics (M and SD) for each group
- Report t statistic and df
- Report p -value
- Report CI immediately after t , e.g., 90% CI [6.156, 9.785]

Directional Hypotheses and One-tailed Tests

- Use directional test only when predicting a specific direction of the difference is justified
- Locate critical region in the appropriate tail
- Report use of one-tailed test explicitly in the research report

Figure 10.6

Two Sample Distributions

Treatment 1	Treatment 2
$n = 9$	$n = 9$
$M = 8$	$M = 13$
$s = 1.22$	$s = 1.22$

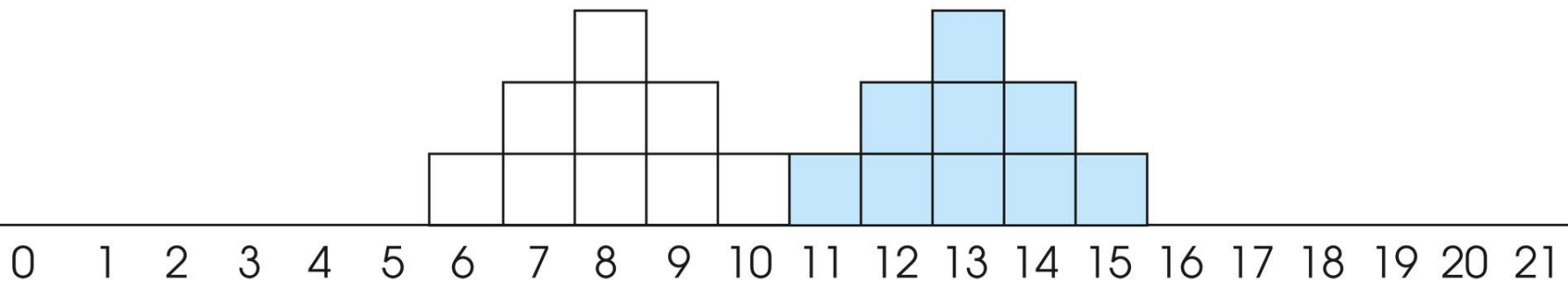


Figure 10.7 Two Samples from Different Treatment Populations

Treatment 1 ☐ Treatment 2 ☐

$$n = 9$$

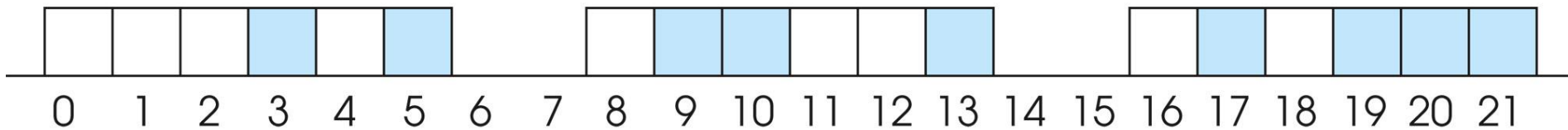
$$M = 8$$

$$s = 6.65$$

$$n = 9$$

$$M = 13$$

$$s = 6.65$$



10.4 Assumptions for the Independent-Measures *t*-Test

- The observations within each sample must be independent
- The two populations from which the samples are selected must be normal
- The two populations from which the samples are selected must have equal variances
 - Homogeneity of variance

Hartley's F -max test

- Test for homogeneity of variance

$$F - \max = \frac{s^2(\text{largest})}{s^2(\text{smallest})}$$

- Large value indicates large difference between sample variance
- Small value (near 1.00) indicates similar sample variances

Box 10.2

Pooled Variance Alternative

- If sample information suggests violation of homogeneity of variance assumption:
- Calculate standard error as in Equation 10.1
- Adjust df for the t test as given below:

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2} \right)^2}{\frac{\left(\frac{s_1^2}{n_1} \right)^2}{n_1 - 1} + \frac{\left(\frac{s_2^2}{n_2} \right)^2}{n_2 - 1}}$$

Learning Check

- For an independent-measures research study, the value of Cohen's d or r^2 helps to describe
-

A

- the risk of a Type I error

B

- the risk of a Type II error

C

- how much difference there is between the two treatments

D

- whether the difference between the two treatments is likely to have occurred by chance

Learning Check

- Decide if each of the following statements is True or False

T/F

- The homogeneity assumption requires the two sample variances to be equal

T/F

- If a researcher reports that $t(6) = 1.98$, $p > .05$, then H_0 was rejected

Chapter 11

The t -Test for Two Related Samples

11.1 Introduction to Repeated-Measures Designs

- Repeated-measures design
 - Also known as within-subjects design
 - Two separate scores are obtained for each individual in the sample
- Same subjects are used in both treatment conditions
- No risk of the participants in each treatment group differing significantly from each other

Matched-Subjects Design

- Approximates the advantages of a repeated-measures design
- Two separate samples are used
 - Each individual in a sample is matched one-to-one with an individual in the other sample.
 - Matched on relevant variables
- Participants are not identical to their match
 - Ensures that the samples are equivalent with respect to some specific variables

Related-Samples Designs

- Related (or correlated) sample designs
 - Repeated-measures
 - Matched samples
- Statistically equivalent methods
- Use different number of subjects
 - Matched sample has twice as many subjects as a repeated-measures design

11.2 t Statistic for Repeated-Measures Research Design

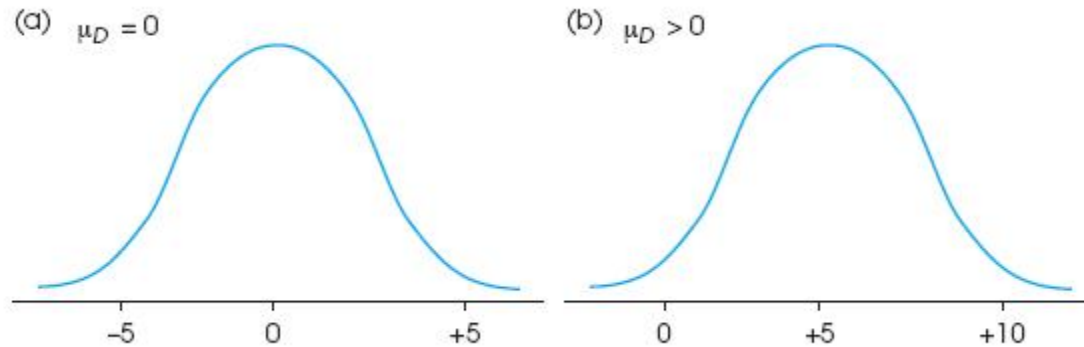
- Structurally similar to the other t statistics
 - Essentially the same as the single-sample t
 - Based on difference scores (D) rather than raw scores (X)
- Difference score = $D = X_2 - X_1$

- Mean Difference
$$M_D = \frac{\sum D}{n}$$

Hypotheses for Related-Samples t Test

- $H_0: \mu_D = 0$
- $H_1: \mu_D \neq 0$

Figure 11.1 Populations of Difference Scores



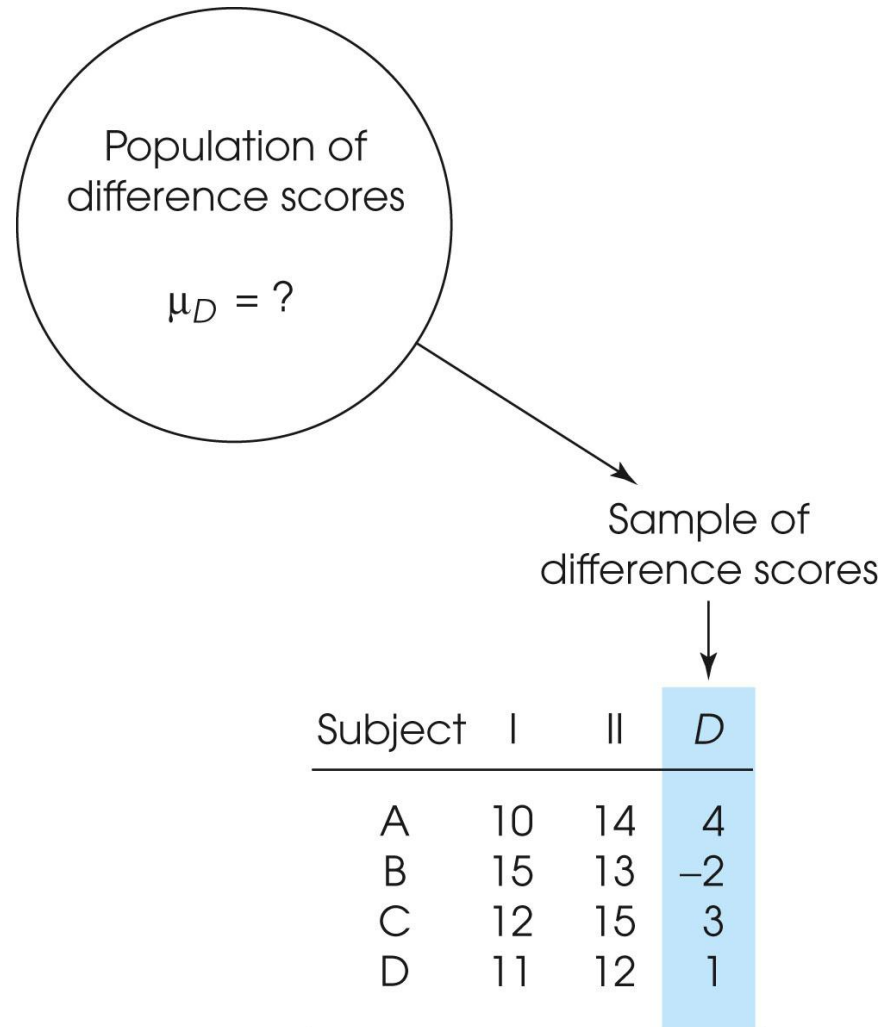
t - Statistic for Related Samples

$$t = \frac{M_D - \mu_D}{s_{M_D}}$$

$$\text{variance of D} = s^2 = \frac{SS}{df}$$

$$s_{M_D} = \sqrt{\frac{s^2}{n}}$$

Figure 11.2 Difference Scores for 4 People Measured Twice



Learning Check

- For which of the following would a repeated-measures study be appropriate?
A matched-subjects study?

A

- A group of twins is tested for IQ

B

- Comparing boys and girls strength at age 3

C

- Evaluating the difference in self-esteem between athletes and non-athletes

D

- Students' knowledge is tested in September and December

Learning Check

- Decide if each of the following statements is True or False

T/F

- A matched-samples study requires only 20 participants to obtain 20 scores in each of the conditions being compared

T/F

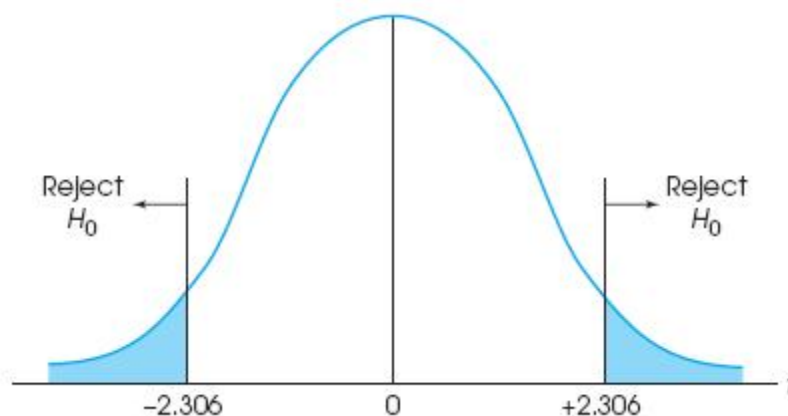
- As the variance of the difference scores increases, the magnitude of the t statistic decreases

11.3 Repeated-Measures Design

Hypothesis Tests and Effect Size

- Numerator of t statistic measures actual difference between the data M_D and the hypothesis μ_D
- Denominator measures the standard difference that is expected if H_0 is true
- Same four-step process as other tests

Figure 11.3 Critical region for t
 $df = 8$ and $\alpha = .05$



Effect size for Related Samples

$$\text{estimated Cohen's } d = \frac{M_D}{s}$$

$$r^2 = \frac{t^2}{t^2 + df}$$

$$C.I.(1-\alpha) : \mu_D = M_D \pm t_{\alpha} s_{M_D}$$

In The Literature

- Report means and standard deviation in a statement or table
- Report a concise version of test results
 - Report t values with df
 - Report significance level
 - Report effect size
- E.g., $t(9) = 2.43$, $p < .05$, $r^2 = .697$

Factors That Influence Hypothesis Test Outcome

- Size of the sample mean difference (larger mean difference → larger numerator so increases t)
- Sample size (larger sample size → smaller standard error—denominator—so larger t)
- Sample variance (larger sample variance → larger standard error—denominator—so smaller t)

Variability as measure of consistency

- When treatment has consistent effect
 - Difference scores cluster together
 - Variability is low
- When treatment effect is inconsistent
 - Difference scores are more scattered
 - Variability is high
- Treatment effect may be significant when variability is low, but not significant when variability is high

Figure 11.4 Example 11.1

Consistent Difference Scores

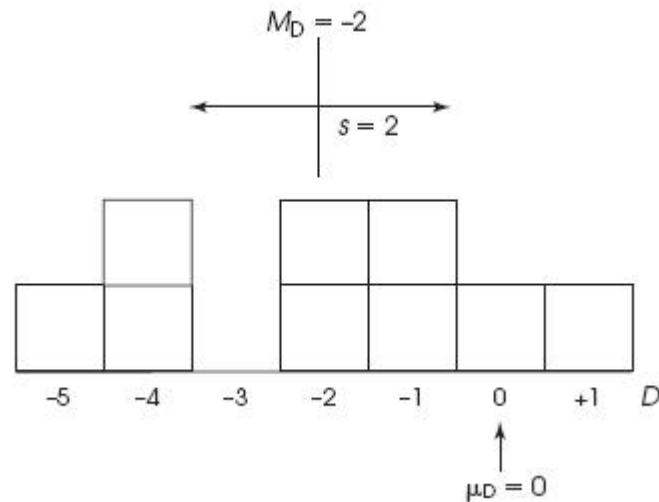
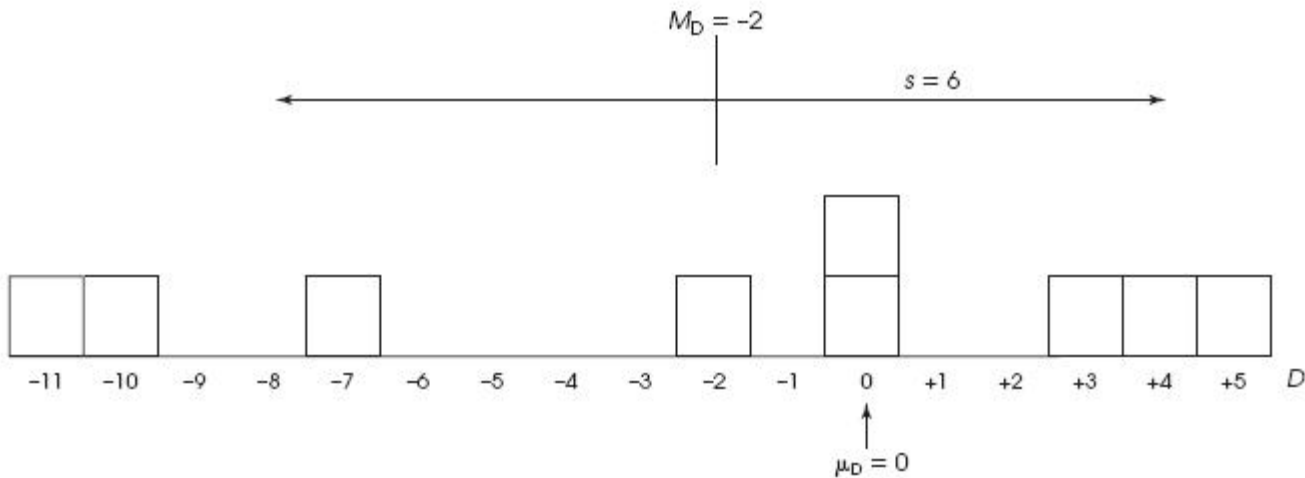


Figure 11.5 A Sample of Inconsistent Difference Scores



Directional Hypotheses and One-Tailed Tests

- Researchers often have specific predictions for related-samples designs
- Null hypothesis and research hypothesis are stated directionally, e.g.
 - $H_0: \mu_D \leq 0$
 - $H_1: \mu_D > 0$
- Critical region is located in one tail

11.4 Related-Samples Vs. Independent-Samples t Tests

- Advantages of repeated-measures design
 - Requires fewer subjects
 - Able to study changes over time
 - Reduces or eliminates influence of individual differences
 - Substantially less variability in scores

11.4 Related-Samples Vs. Independent-Samples t Tests

- Disadvantages of repeated-measures design
 - Factors besides treatment may cause subject's score to change during the time between measurements
 - Participation in first treatment may influence score in the second treatment (order effects)
- Counterbalancing is a way to control time-related or order effects

Related-Samples t Test

Assumptions

- Observations within each treatment condition must be independent
- Population distribution of difference scores (D values) must be normally distributed
 - This assumption is not typically a serious concern unless the sample size is small.
 - With relatively large samples ($n > 30$) this assumption can be ignored

Learning Check

- Assuming that the sample mean difference remains the same, which of the following sets of data is most likely to produce a significant t statistic?

A • $n = 15$ and $SS = 10$

B • $n = 15$ and $SS = 100$

C • $n = 30$ and $SS = 10$

D • $n = 30$ and $SS = 100$

Learning Check

- Decide if each of the following statements is True or False

T/F

- Compared to independent-measures designs, repeated-measures studies reduce the variance by removing individual differences

T/F

- The repeated-measures t statistic can be used with either a repeated-measures or a matched-subjects design

Homework

- Chapter 10
- Q6 Q8 Q14
- Chapter 11
- Q4 Q10